

A Novel Telephony Recognizer for Dysarthric Speech

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Motivation

-  Single-switch users
 -  Users w/ severe mobility restrictions (ex. quadriplegics)^[1]
 -  Can only create a single, consistent movement^[1]
 -  Cannot use touch interfaces
 -  Use single-switch devices and scanning (a system which automatically iterates through interface options)
-  My single-switch device (for emulation): Tecla
-  May have comorbid conditions such as dysarthria (motor speech disorder)

Basic Introduction to Switch Control



Research Problem

-  Phone call functions are difficult for single-switch users
-  Telephony functions must be available offline
-  The Time-Sensitive Telephony Problem:
 1. Complex menu systems cause never-ending loops
 -  Too many options to parse
 -  Users wait for the option but passes too quickly
 2. Calls automatically go to voicemail
 -  Ex. Fido sets max at 6 rings/30 seconds^[2]

Slow & Steady Doesn't Win The Race

DYSARTHRIC SPEECH

≤ 15

words per
minute^[3]

NON-DISORDERED SPEECH

160-200

words per minute
on average^[4]

SINGLE-SWITCH TYPING RATES

2-4

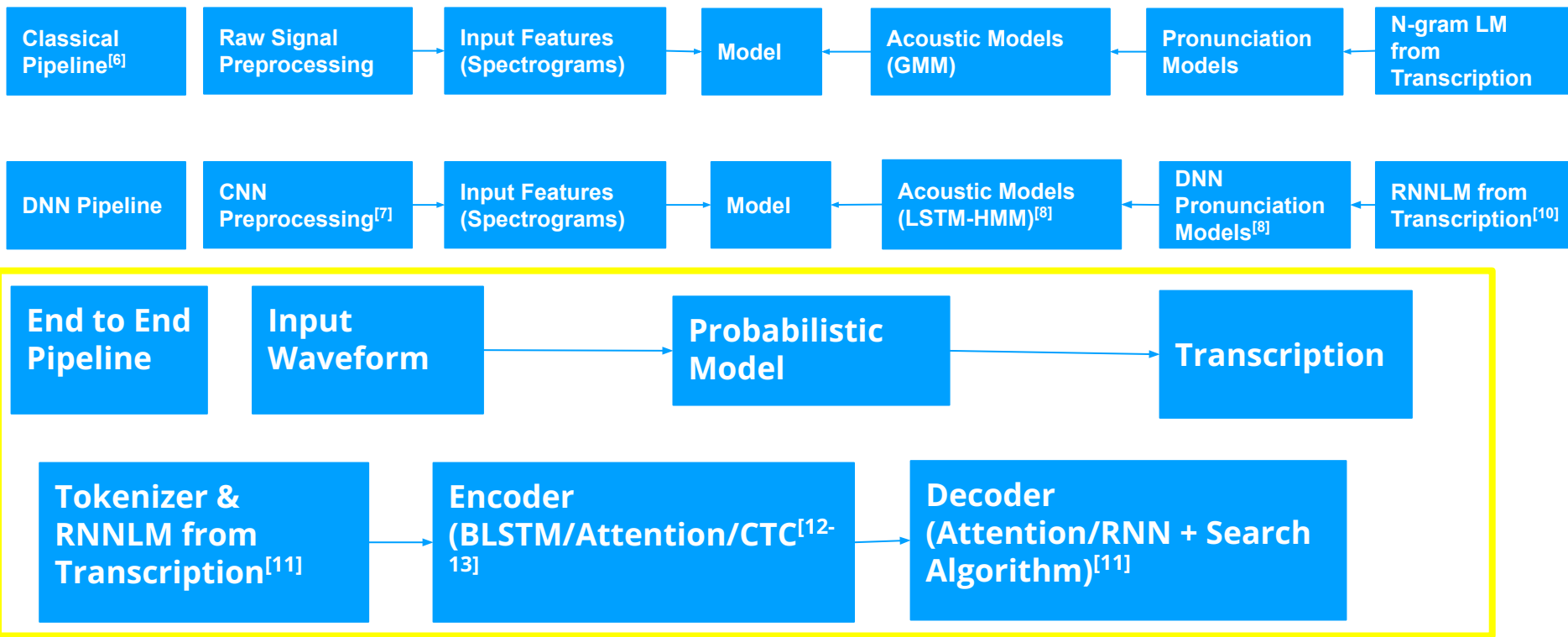
words per
minute for
typical touch
interfaces^[5]

TOUCH TYPING RATES

20-45

words per
minute, with
word
prediction^[5]

Literature Review: The Automatic Speech Recognition (ASR) Pipeline



Literature Review: Dysarthric ASR Systems

	STARDUST[11]	PYKALDI & DEEPSPEECH [15,16]	SIRI, ALEXA & DRAGON[20,21]	GOOGLE PROJECT RELE	EASYCALL[19]	TORGO[3] & UASPEECH[17]	HOMESERVICE[18]
USES MODERN DNNS							
LIMITED VOCABULARY							
LENGTHY IMPLEMENTATION							
SUFFICIENT DATA							
OPEN SOURCE							
ENGLISH SUPPORT							
EXTENSIBLE							
SPEECH IMEDPIMENT COMPATIBLE							

Research Questions

1. What are the factors influencing recognition of dysarthric speech and limited-vocabulary language models in offline recognizers?
2. How can ASR and DNNs specific to dysarthric speech be combined to increase the selection efficiency and user satisfaction for access to smartphone telephony functions for single switch users?

Methodology



Design a limited vocabulary ASR system rather than a LVCSR



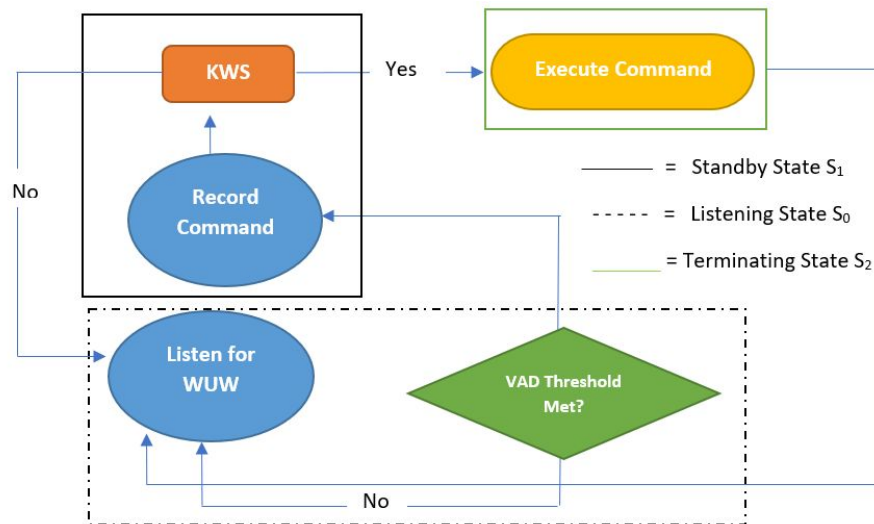
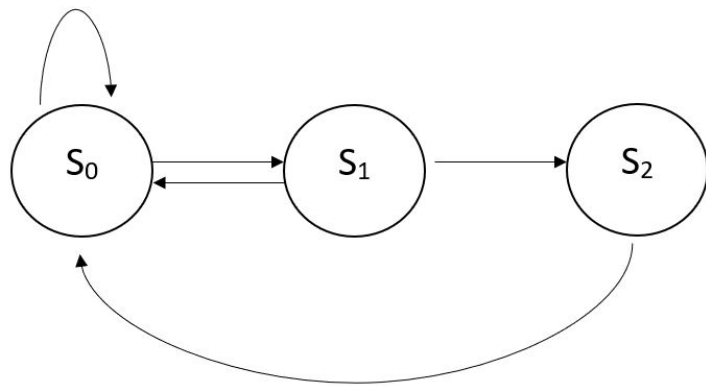
Use a two-stage training and WWD process



Train a general Deep Neural Network (DNN) model on dysarthric speech



Fine-tune to a Keyword Spotter (KWS)



Data Collection

 Rule of Thumb: DNN training requires ~100,000 data points^[23-25]

 Dysarthric Data: 50-75% of required data (omitting controls)

Dataset	Hours	Observations
TORG ^[3]	15	~6000
UASpeech ^[17]	102.7	45,465
Nemours ^[26]	<3	814
EasyCall ^[19]	-	11,309
homeService ^[18]	10	1286
Sum of Dysarthric	~150	~65,000
LibriSpeech (non-disordered comparison) ^[27]	1000	~ 290,000

A single non-disordered dataset surpasses this minimum requirement, creating a divide b/w recognition of disordered and non-disordered speech



The *SpeechBrain* of the Operation



A PyTorch-based all-in-one speech toolkit^[11]



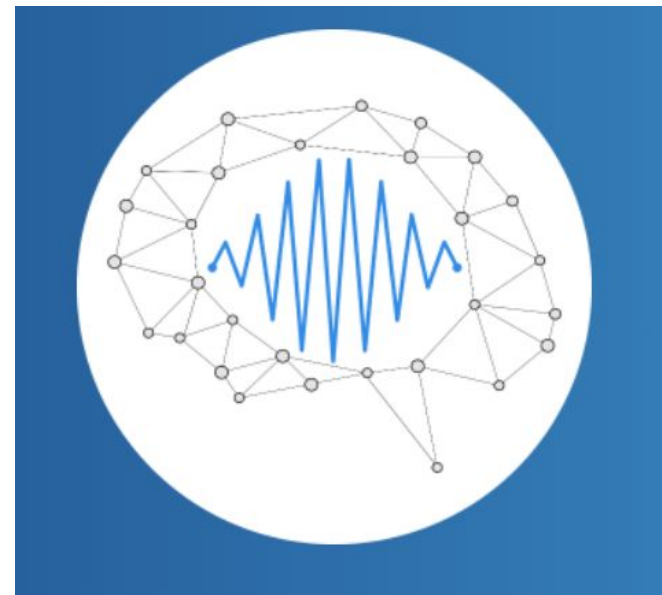
Allows modular configurations of various DNNs and Transformers for a fully customizable E2E system



Training on the 5 dysarthric speech datasets is in progress



Supports ASR and KWS



Credit <https://speechbrain.github.io/>

Model Training

- Encoder: Convolutional Recurrent Deep Neural Network (CRDNN)^[11]
 - RNN can be Long Short Term Memory (LSTM), Gated Recurrent Unit (GRU) or Light-GRU (Li-GRU)
 - Loss: Connectionist Temporal Classification (CTC) + Negative Log Likelihood (NLL)
 - Google's SentencePiece Tokenizer determines lexical breakdown
- Decoder: GRU + Beamsearch + RNNLM^[12]
- Attention Transformer can replace Encoder + Decoder and combined with CTC
 - Research indicates combination is worth exploring^[28-35]

Discussion

-  Original goal: Fine tune a KWS from a LV dysarthric speech model
-  Amended goal: Use string match on transcription and skip fine-tuning
-  KWS requires sufficient data to contain keywords
-  Data Imbalance: Only ~10% of data is appropriate for telephony commands
-  Request users for relevant recordings? Issue remains
-  Data can be synthesized but dealing w/ such a large proportion of synthesized data is out-of-scope
 -  Ex. Google's SpecAugment creates variations on spectrograms of data points to artificially create data^[36]

Next Steps

-  Use ASR without WWD and string-match transcriptions of voiced samples to expected command
-  Experiment w/ different neural architectures
-  iOS proof-of-concept asks users to record command and simulates call
 -  70% of Tecla's consumer base uses iOS, so the iOS platform is appropriate for this population
-  Ethics Application
-  User Evaluation
 -  User's ability to interact with the app using a single-switch
 -  Recognition performance of spoken commands

Research Challenges

1. Cannot Train Model

 Issues w/ neuron ratio, padding, etc.

2. Pre and post questionnaire tools (ex. Google Forms, Qualtrics, etc.) are not single-switch accessible

 Suggested workaround is verbal yes/no interview

3. String match feasible alternative to KWS?

 Assign a 'None' class?

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Thank you for your time!



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